AMCG-3: UNIFIED ANTI-COMPACTION DIGEST & 3-GEAR MULTIMODAL VERIFICATION ENGINE

Eliminating Auto-Compaction Amnesia, Cascading Death Spirals, Hook Gaming & Unverified Agentic Outputs Across Multi-Agent Pipelines

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LAYER 1 · INGESTION

Universal Multi-Agent Adapter

· Normalizes lifecycle hooks across Antigravity, Claude Code, Gemini CLI, Cursor & Codex Proxy.
· Reconciles Undo/Rewind log branches between rewritten transcript_full.jsonl and append-only compact JSONL trajectory logs.
· Prevents truncated_fields: [tool_calls] blindness on large parallel tool-call batches.
· Exempts browser subagent # Browser Session Checkpoint markers from parent digest locks.

LAYER 2 · MEMORY CONTINUITY

UTF-8 Byte-Bounded Digest Synthesizer

· Distills 10.6 MB raw JSONL into <= 56.4 KB post_compaction_digest.md (<= 345 lines).
· Section 1: 100% Verbatim Chronological User Prompts from Step 0 -> Latest Step.
· Section 2: Canonical Turn-by-Turn Tool Calls, File Edits, Tool Errors & Model Replies.
· 3-byte Bangla/Unicode line bounding guarantees <= 22.0 KB per 120-line chunk (0% truncation).

LAYER 3 · ADVERSARIAL FIREWALL

Cryptographic Anti-Bypass Hook Lock

· Hard-blocks ByteOffset > 0 overflow exploits (caught live in trajectory 25f507a4 Step 1382).
· Blocks 5-line micro-read probing; enforces 100% line-by-line coverage [1..N_L].
· Enforces strict Conversation-UUID path isolation (blocks cross-session spoofing).
· Header-only error detection prevents digest-body historical error text collisions.

LAYER 4 · ADAPTIVE ROUTING

3-Gear Risk & Cost Controller

· Gear 1 (Fast / Default): Post-Compaction Digest + Binary Crash Guard + Dead-Link Check.
· Gear 2 (Standard / Web & Code): + Automated Playwright Dual-Viewport (1440x900 & 390x844) UI Screenshot & DOM Console Error Gate.
· Gear 3 (Critical / Media): + Full 4-Tier Strict Liability (Worker -> Supervisor -> Chief Auditor -> Tier-0 SI) & 4-File Video QC.

-98.1% to -99.6%
Raw JSONL Bloat Reduction (10.6 MB -> 49.8 KB)

0.0% Loss
46 KB Tool Byte-Truncation (Max Chunk 22.0 KB)

0 Loops
Cascading Re-Compaction Death Spirals Eliminated

6 / 6 Blocked
Adversarial Hook-Gaming Exploits Neutralized

11 Trajectories
9,165 Production Steps Forensically Audited

MODULE 1: Media & Lip-Sync Multimodal Gate

GEAR 3 · PyAV + OpenCV + Faster-Whisper

· Signal Physics: Hard-clamped frozen-frame (<= 1.8s) and dead-silence (<= 1.35s) detectors plus 40ms RMS dB cut-boundary speech clipping verification.
· AST/Code Anti-Shortcut Scan: Rejects FFmpeg reverse boomerang padding, anullsrc (>= 0.65s) synthetic gaps, and sine-wave camera wobble cheats.
· 4-File Multimodal Proof Bundle: Generates 1s full-frame contact sheet, 0.5s close-up mouth/energy grid, 360p preview MP4, and master MP3 audio.
· Mathematical Invariant: E_RMS(t_cut +/- 40ms) <= -22.0 dB across all cuts.

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Receipt Lock: mtime + byte-size verified across 4-Tier Chain
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MODULE 2: Web & UI Visual Multimodal Gate

GEAR 2 · Headless Chromium + DOM Geometry

· Dual-Viewport Rendering: Automatically renders local servers or HTML builds at Desktop (1440x900) and Mobile (390x844) via Playwright Chromium.
· Zero Runtime Defect Gate: Captures uncaught pageerror, console.error, failed network assets, and broken images (naturalWidth == 0).
· Layout Geometry Check: Detects horizontal viewport overflow (scrollWidth > innerWidth) and produces <150 KB JPEG visual proof sheets.
· Eliminates blind 'UI Complete' claims without visual & DOM console proof.

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Enforced via web_ui_qc_gate.py -> web_ui_qc_receipts.json
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MODULE 3: Crash Firewall & Hermes Lifelong Memory

GEAR 1-3 · PyMuPDF + PIL + Self-Evolving Ledger

· Trajectory Poisoning Guard: Blocks read_file on password-encrypted PDFs, fake-extension HTML/0-byte media, and >450 KB raw images before HTTP 400 crash.
· Post-Edit Practical Run Gate: Blocks Stop hook whenever source code (.py, .ts, .js, .ps1) was edited without subsequent practical command verification.
· Hermes Memory Engine: Harvests solved bugs into hermes_ledger.json with regex pre-tool firewalls so historical mistakes never recur across chats.
· Cryptographic State Machine: F_allow(t, C_k) = 1 iff Cov(R_valid) = [1..N_L].

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Permanent Cross-Session Immunity Across All Agentic Tools
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EMPIRICAL TELEMETRY BENCHMARK ACROSS 11 PRODUCTION TRAJECTORIES (VF-AI-MEM-2026-005 / DOI: 10.5281/zenodo.24096896 & 10.5281/zenodo.24097022):

· e2cdad79 (1,841 steps, 10.63 MB raw -> 49.8 KB digest, -99.5% bloat, 21.8 KB max 120-line chunk, 0% truncation) · c8bf1bf6 (679 steps, 7 compactions resolved -> 36.2 KB digest, -98.8%)
· 25f507a4 (1,622 steps, 5.91 MB raw -> 56.4 KB digest, -99.0% bloat, ByteOffset=5000000 exploit blocked) · 4bcf6d81 (1,295 steps, 3.84 MB raw -> 31.4 KB digest, -99.2% bloat)